Ques 1:-

Given,

(c)

$C/W_x = 20$	$C/W_y = 15$
$D_x = 19\,cm$	$D_y = 20\,cm$
$R_x = \frac{19}{2}\,cm$	$R_y = \frac{20}{2}\,cm$
$D/W_x = 89$	$D/W_y = 100$
$Def/A_x = 0.023$ defects/cm²	$Def/A_y = 0.031$ defects/cm²

Use the updated formula for yield to solve this question

$$Yield = \frac{1}{(1+(Defects\ per\ area \times Die\ area))^N}$$ **Note:** N will be given.

$$Yield_x = \frac{1}{\left(1+\left(Defects\ per\ area \times \frac{Die\ Area}{2}\right)\right)^2}$$

$$= \frac{1}{\left(1+\left(0.023 \times \frac{Die\ Area}{2}\right)\right)^2}$$

$$= \frac{1}{\left(1+\left(0.023 \times \frac{3.1857}{2}\right)\right)^2}$$

$$\cong 0.93$$

$$D/W = \frac{Wafer\ Area}{Die\ Area}$$

$$Die\ Area = \frac{Wafer\ Area}{D/W}$$

$$= \frac{\pi \times \left(\frac{19}{2}\right)^2}{89}$$

$$= 3.1857$$

$$Yield_y = \frac{1}{\left(1+\left(Defects\ per\ area \times \frac{Die\ Area}{2}\right)\right)^2}$$

$$= \frac{1}{\left(1+\left(0.031 \times \frac{Die\ Area}{2}\right)\right)^2}$$

$$= \frac{1}{\left(1+\left(0.031 \times \frac{3.1416}{2}\right)\right)^2}$$

$$\cong 0.91$$

$$D/W = \frac{Wafer\ Area}{Die\ Area}$$

$$Die\ Area = \frac{Wafer\ Area}{D/W}$$

$$= \frac{\pi \times \left(\frac{20}{2}\right)^2}{100}$$

$$= 3.1416$$

d) $$C/D_x = \frac{Cost\ per\ wafer}{Dies\ per\ wafer \times Yield}$$

$$= \frac{20}{89 \times 0.93}$$

$$= 0.24$$

$$C/D_y = \frac{Cost\ per\ wafer}{Dies\ per\ wafer \times Yield}$$

$$= \frac{15}{100 \times 0.91}$$

$$= 0.16$$

Ques 2:

	A	B	C	D
CPI_{PS}	7	2	3	6
CPI_{XB}	5	4	2	1
IC	$1\times10^6\times.3 = 3\times10^5$	$1\times10^6\times.5 = 5\times10^5$	$1\times10^6\times.1 = 1\times10^5$	$1\times10^6\times.1 = 1\times10^5$

$Rate_{PS} = 2.7\,GHz = 2.7\times10^9\,Hz$

$Rate_{XB} = 3\,GHz = 3\times10^9\,Hz$

$Instruction\ Count = 1\times10^6$

a) For Play Station,

$$\text{Clock Cycles} = (7\times3\times10^5 + 2\times5\times10^5 + 3\times1\times10^5 + 6\times1\times10^5$$

$$= 4\times10^6$$

$$\text{Avg. CPI} = \frac{4\times10^6}{1\times10^6}$$

$$= 4$$

For XBOX,

$$\text{Clock Cycles} = (5\times3\times10^5 + 4\times5\times10^5 + 2\times1\times10^5 + 1\times1\times10^5$$

$$= 3.80\times10^6$$

$$\text{Avg. CPI} = \frac{3.80\times10^6}{1\times10^6}$$

$$= 3.8$$

So, Difference $= (4 - 3.8)$

$= 0.2$

b) $$\text{Ex time}_{PS} = \frac{\text{Instruction Count} \times CPI}{\text{Rate}}$$

$$= \frac{10^6\times4}{2.7\times10^9}$$

$$= 0.00148\ s$$

$$\text{Ex time}_{PS} = \frac{\text{Instruction Count} \times CPI}{\text{Rate}}$$

$$= \frac{10^6\times3.8}{3\times10^9}$$

$$= 0.00126\ s$$

Difference $= (0.00148 - 0.00126)$

$= 0.22$

Execution time of a Reference processor

Reference time is provided by SPEC

Measured time is the Execution time of a given

c) $$\text{Spec Ratio} = \frac{\text{Reference Time}}{\text{Measured Time}}$$

$$= \frac{120}{\text{Ins. Count} \times CPI \times C.eT}$$

$$= \frac{120\times10^{-3}}{10^6\times4\times\frac{1}{2.7\times10^9}}$$

$= 81$

Ques-3

Processor	CPI	Rate
P1	1.5	3 GHz
P2	1.0	2.5 GHz
P3	2.2	4 GHz

Instruction Count = X
[same for all Processors]

a) $\text{Performance (IPS)} = \frac{\text{Clock Rate}}{\text{CPI}}$

$P_{P1} = \frac{3 \times 10^9}{1.5} = 2 \times 10^9$

$P_{P2} = \frac{2.5 \times 10^9}{1} = 2.5 \times 10^9$ ⇒ highest perf.

$P_{P3} = \frac{4 \times 10^9}{2.2} \cong 1.82 \times 10^9$

Rate = cycles completed per second

for P1,

3×10^9 cycles --- 1 s

∴ 1 " ... $\frac{1}{3 \times 10^9}$ s

∴ 1.5 " ... $\frac{1.5}{3 \times 10^9}$ s

[1 ins = 1.5 cycle]

So,

1 ins. takes $\frac{1.5}{3 \times 10^9}$ s to complete

$\frac{1.5}{3 \times 10^9}$ s to complete – 1 ins

∴ 1 s " " $\frac{3 \times 10^9}{1.5}$ ins (Rate / CPI)

$\text{Performance (IPS)} = \frac{\text{Clock Rate}}{\text{CPI}}$

b) $CPU_{Time} = 10s$

$CPU_{time} = \frac{IC \times C.P.I}{freq}$

$\Rightarrow IC = \frac{CPU_{time} \times freq}{C.P.I}$

$IC(P1) = \frac{10 \times 3 \times 10^9}{1.5} = 2 \times 10^{10}$

$IC(P2) = \frac{10 \times 2.5 \times 10^9}{1.0} = 2.5 \times 10^{10}$

$IC(P3) = \frac{10 \times 4 \times 10^9}{2.2} = 1.8 \times 10^{10}$

$CPU_{time} = \frac{CPU_{clock\ cycles}}{Freq} = \frac{IC \times CPI}{Freq}$

$\Rightarrow CPU_{clock\ cycles} = IC \times CPI$

∴ Clock Cycles (P_1) = $2 \times 10^{10} \times 1.5 = 3 \times 10^{10}$

∴ Clock Cycles (P_2) = $2.5 \times 10^{10} \times 1 = 2.5 \times 10^{10}$

∴ Clock Cycles (P) = $1.8 \times 10^{10} \times 2.2 \cong 3.96 \times 10^{10}$

the more digit you take after fraction the less error you get!

c)

Processor	CPI	New CPI
P1	1.5	1.8
P2	1.0	1.2
P3	2.2	2.64

$$CPU\ Time_{new} = 10 - (10 \times .3)$$
$$= 7s$$

$$CPU_{Time} = \frac{I.C \times CPI}{Freq.}$$

$$\Rightarrow Freq. = \frac{IC \times CPI}{CPU_{Time}}$$

$$F(P_1) = \frac{IC \times CPI}{CPU_{Time}} = \frac{2 \times 10^{10} \times 1.8}{7} \cong 5.14\ GHz$$

$$F(P_2) = \frac{IC \times CPI}{CPU_{Time}} = \frac{2.5 \times 10^{10} \times 1.2}{7} \cong 4.28\ GHz$$

$$F(P_3) = \frac{IC \times CPI}{CPU_{Time}} = \frac{1.8 \times 10^{10} \times 2.64}{7} \cong 6.79\ GHz$$

Quen-4 :

$$CPU_{Time} = 250\ s$$

Add	Sub	left-Shift
70 s	85 s	40 s

$= 105s$

mine. $\Rightarrow (250 - 105) = 55$

a) $T_{new} = (70 \times 0.8) + 85 + 40 + 55$

$= 236s$

time reduced $= (250 - 236) = 14s$

$= \frac{14}{250} \times 100\% \cong 5.47\%$

b) $T_{new} = (250 \times 0.8) = 200s$

$T_{add} + T_{sub} + T_{mine} = 70 + 85 + 55 = 210s > T_{new}.$

NO

Ques 5:

$$MIPS = \frac{\text{Clock Rate}}{CPI \times 10^6}$$

Rate $(P_1) = 4\,GHz = 4 \times 10^9\,Hz$

$CPI\,(P_1) = 0.9$

Ins. Count $= 5 \times 10^9$

$$MIPS(P_1) = \frac{4 \times 10^9}{0.9 \times 10^6}$$

$$= 4.44 \times 10^3$$

Rate $(P_2) = 3\,GHz = 3 \times 10^9\,Hz$

$CPI\,(P_2) = 0.75$

Ins. Count $= 1 \times 10^9$

$$MIPS(P_2) = \frac{3 \times 10^9}{0.75 \times 10^6}$$

$$= 4 \times 10^3$$

$$P(P_1) = \frac{4 \times 10^9}{5 \times 10^9 \times 0.9} = 0.889$$

$$P(P_2) = \frac{3 \times 10^9}{1 \times 10^9 \times 0.75} = 4$$

$MIPS(P_1) > MIPS(P_2)$

$Perf.(P_1) < Perf.(P_2)$

False

Ques -6

$$power = CL \times V^2 \times F$$

$$\frac{P_{new}}{P_{old}} = \frac{C_{old} \times 0.85 \times (V \times 0.81)^2 \times F \times 0.81}{C_{old} \times V^2 \times F}$$

$$= \frac{C_{old} \times 0.85 \times V^2 \times 0.81^2 \times F \times 0.81}{C_{old} \times V^2 \times F}$$

$$= 0.4605$$

about

The new processor uses almost Half (0.4605) power of the old processor.

Ques - 7

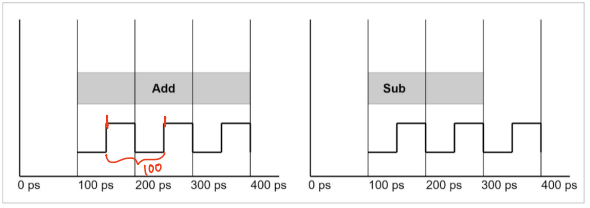


Figure 1: Represents 1 add instruction sub instruction

Figure 2: Represents 1 instruction

Figure 1: Represents 1 add instruction

Figure 2: Represents 1 sub instruction

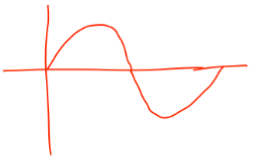
Program **A** is divided into two classes according to their **CPI** (Add and Sub). The **instruction counts** are 21 and 3 respectively. Reference for **program A** is 1080ps.
Now, answer the following questions,

a) What is the Clock period? **Hint**: follow any of the figures
b) What is the frequency?
[1]
c) What is the CPI for Add and Sub ?
[2]
d) What is the Avg. CPI?
[2]
e) Find out the execution time of the program?
[2]
f) Find the SPEC ratio?
[1]
g) If you want to improve the performance by 1.2 times, what improvement do you need to include in the program's add operation?
[2]

Program A

Class	Add	Sub
CPI	3	2
IC	21	3

$\frac{(21\times3)+(3\times2)}{24} = 2.875$





Duration between two rising edges = 100 ps = 100×10^{-12} s

Freq = $\frac{1}{100\times10^{-12}} = 10^{10}$ Hz $= 10\times10^{9}$ Hz $= 10$ GHz

Clock cycle per ins. → 3, 2

2.875

Clock Cycle

$\Rightarrow$ Ex CPU Time = Ins./Prog × CPI × Clock Period

$= 24 \times 2.875 \times 100\times10^{-12}$

$= 6.9\times10^{-9}$ s / 6900 ps

Spec Ratio = $\frac{\text{Ref. time}}{\text{CPU time / Execution time}}$

g)

$\text{Time}_{\text{improved}} = \frac{T_{\text{affected}}}{n} + T_{\text{unaff.}}$

$\Rightarrow \frac{6.9\times10^{-9}}{1.2} = \frac{6.0375\times10^{-9}}{n} + 8.625\times10^{-10}$

$\Rightarrow n = 1.235$

$= \frac{1080 \text{ ps}}{6900 \text{ ps}}$

$= 0.1565$

$T_{\text{aff.}} = 21 \times 2.875 \times 100\times10^{-12}$

$= 6.0375\times10^{-9}$